

PROJECT TASK – 2

BUILD YOUR PERSONAL CYBERSECURITY LAB

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Date:

22 June 2026

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1. Objective

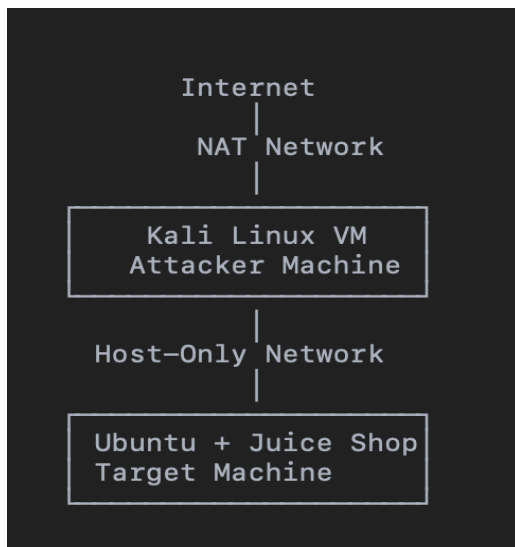
The objective of this project was to build a secure and isolated cybersecurity lab using virtualization. The lab consists of an attacker machine (Kali Linux) and a vulnerable target machine (Ubuntu running OWASP Juice Shop). The environment was configured using NAT and Host-Only networking to safely perform penetration testing activities without affecting real-world systems.

2. Software Used

Software	Purpose
VMware Fusion	Virtualization
Kali Linux	Attacker Machine
Ubuntu 24.04	Target Machine

Docker	Deploy Juice Shop
OWASP Juice Shop	Vulnerable Web Application
Nmap	Network Scanning
Burp Suite	Web Proxy
Wireshark	Packet Capture

3. Lab Architecture

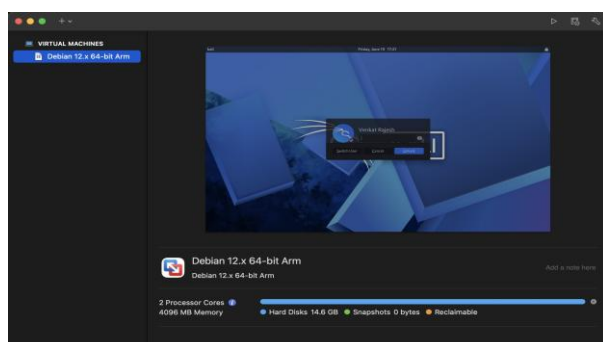


4. VMware Configuration

VMware Fusion was used to create two virtual machines. Both machines were configured with two network adapters:

- Adapter 1: NAT (Internet access)
- Adapter 2: Host-Only (Private communication between VMs)

Screenshots:





5. Kali Linux Configuration

Kali Linux was configured as the attacker machine. After enabling NAT and Host-Only networking, IP addresses were verified using the `ip a` command.

Screenshots:

```

Session  Actions  Edit  View  Help
┌─(venkat@kali)-[~]
│$ ip a
│1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
│   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
│   inet 127.0.0.1/8 scope host lo
│       valid_lft forever preferred_lft forever
│   inet6 ::1/128 scope host noprefixroute
│       valid_lft forever preferred_lft forever
│2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
│   link/ether 00:0c:29:ec:e9:67 brd ff:ff:ff:ff:ff:ff
│   inet 192.168.161.129/24 brd 192.168.161.255 scope global dynamic noprefixroute eth0
│       valid_lft 1771sec preferred_lft 1771sec
│   inet6 fe80::20c:29ff:feec:e967/64 scope link noprefixroute
│       valid_lft forever preferred_lft forever
│3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
│   link/ether 00:0c:29:ec:e9:71 brd ff:ff:ff:ff:ff:ff
│
┌─(venkat@kali)-[~]
│$ 
└─

```

6. Ubuntu Configuration

Ubuntu was installed as the vulnerable target system. Docker was installed and configured to deploy the OWASP Juice Shop application.

Screenshots:



```

ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:55:32:b2 brd ff:ff:ff:ff:ff:ff
    altname enp2s0
    altname enx000c295532b2
    inet 192.168.161.130/24 brd 192.168.161.255 scope global dynamic noprefixroute ens160
        valid_lft 1592sec preferred_lft 1592sec
    inet6 fe80::20c:29ff:fe55:32b2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens192: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:55:32:bc brd ff:ff:ff:ff:ff:ff
    altname enp10s0
    altname enx000c295532bc
    inet 172.16.64.128/24 brd 172.16.64.255 scope global dynamic noprefixroute ens192
        valid_lft 1592sec preferred_lft 1592sec
    inet6 fe80::8ee1:adf0:d94b:d4e9/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$

```

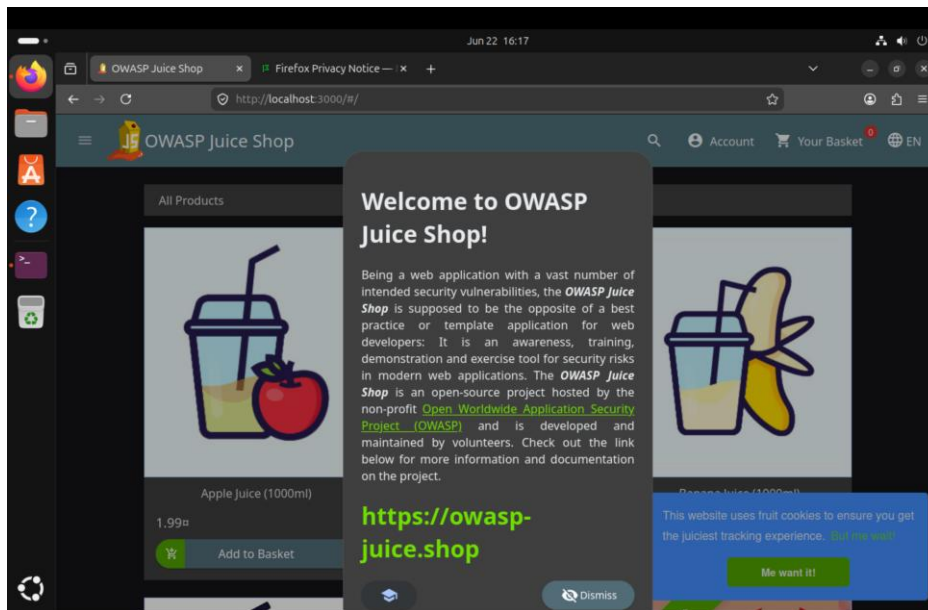
7. Docker Installation

Docker was installed on Ubuntu using the APT package manager. The Docker service was started successfully, and the Juice Shop Docker container was launched.

8. OWASP Juice Shop

OWASP Juice Shop was deployed inside a Docker container and accessed successfully through the browser using port 3000.

Screenshots:



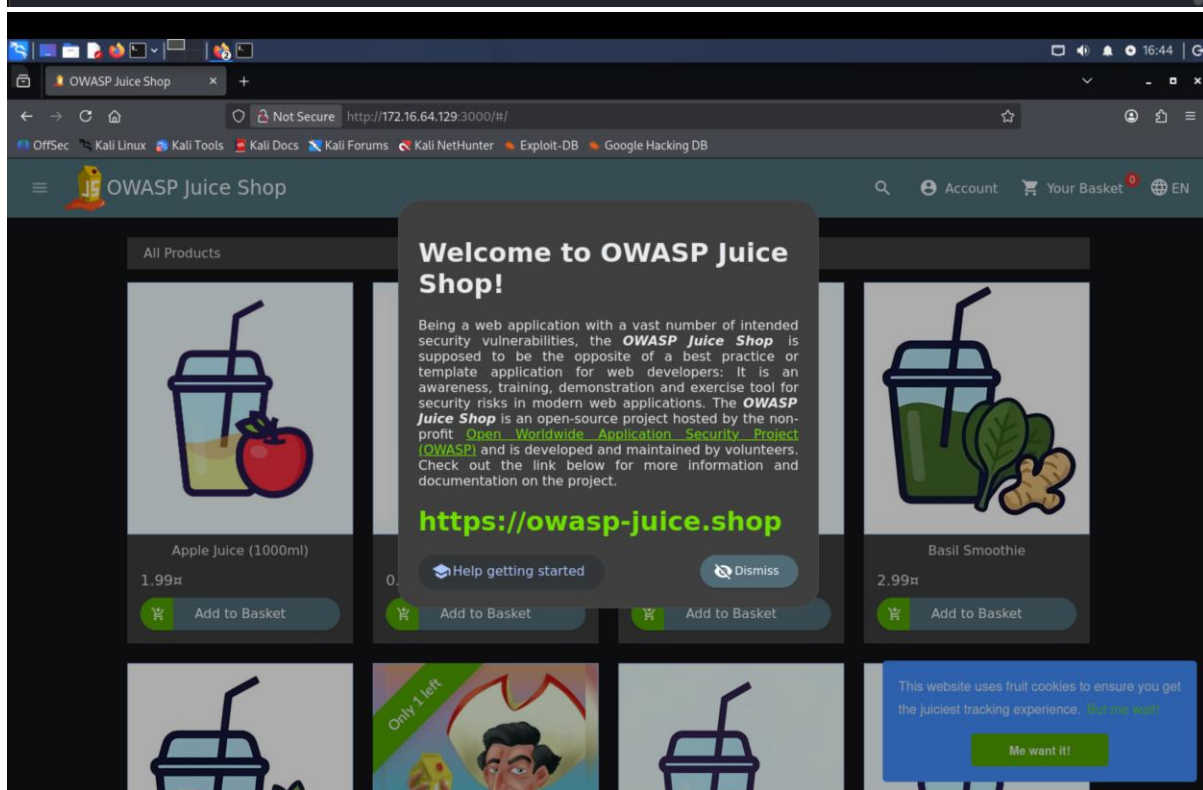
9. Network Validation

Connectivity between Kali Linux and Ubuntu was verified using ICMP ping over the Host-Only network.

Screenshots:

```
(venkat@kali)-[~]
$ ping 172.16.64.129
PING 172.16.64.129 (172.16.64.129) 56(84) bytes of data.
64 bytes from 172.16.64.129: icmp_seq=1 ttl=64 time=0.634 ms
64 bytes from 172.16.64.129: icmp_seq=2 ttl=64 time=0.566 ms
64 bytes from 172.16.64.129: icmp_seq=3 ttl=64 time=0.564 ms
64 bytes from 172.16.64.129: icmp_seq=4 ttl=64 time=0.549 ms
64 bytes from 172.16.64.129: icmp_seq=5 ttl=64 time=0.530 ms
64 bytes from 172.16.64.129: icmp_seq=6 ttl=64 time=0.536 ms
64 bytes from 172.16.64.129: icmp_seq=7 ttl=64 time=0.384 ms
64 bytes from 172.16.64.129: icmp_seq=8 ttl=64 time=0.622 ms
64 bytes from 172.16.64.129: icmp_seq=9 ttl=64 time=0.560 ms
64 bytes from 172.16.64.129: icmp_seq=10 ttl=64 time=0.564 ms
64 bytes from 172.16.64.129: icmp_seq=11 ttl=64 time=0.575 ms
64 bytes from 172.16.64.129: icmp_seq=12 ttl=64 time=0.584 ms
64 bytes from 172.16.64.129: icmp_seq=13 ttl=64 time=0.568 ms
^C
— 172.16.64.129 ping statistics —
13 packets transmitted, 13 received, 0% packet loss, time 12282ms
rtt min/avg/max/mdev = 0.384/0.556/0.634/0.057 ms

(venkat@kali)-[~]
```



10. Nmap Scan

Nmap was used to scan the target machine and identify open ports and running services. The scan confirmed that the web application was accessible.

Screenshots:

```
(venkat@kali)-[~]
$ nmap -sV 172.16.64.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-22 16:46 +0530

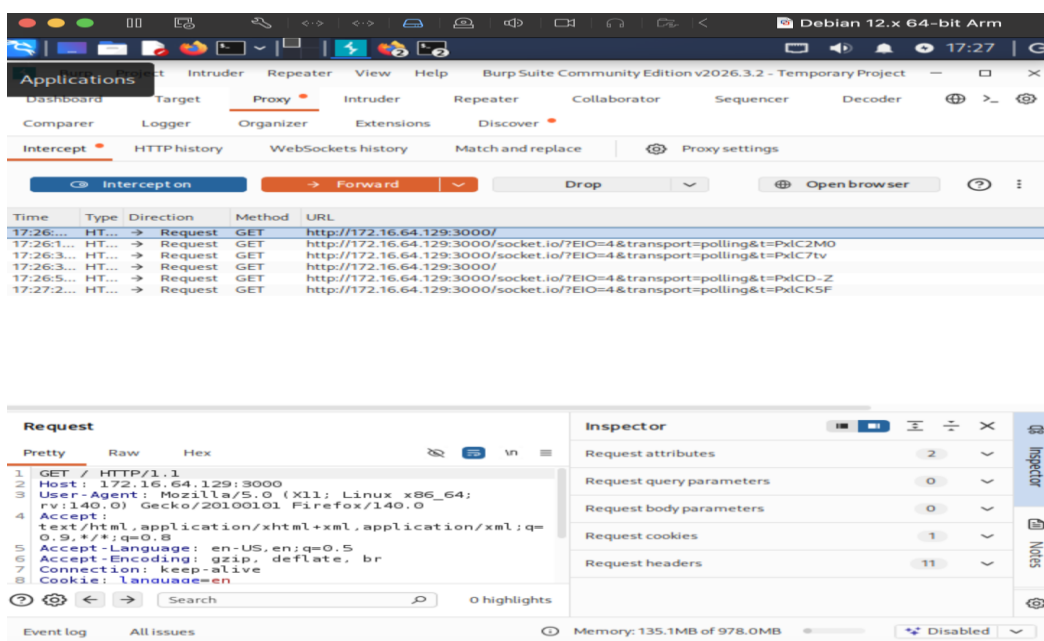
(venkat@kali)-[~]
$ nmap -sV 172.16.64.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-22 16:47 +0530

(venkat@kali)-[~]
$ nmap 172.16.64.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-22 16:48 +0530
```

11. Burp Suite

Burp Suite intercepted HTTP requests between the browser and OWASP Juice Shop, demonstrating successful proxy configuration and traffic inspection.

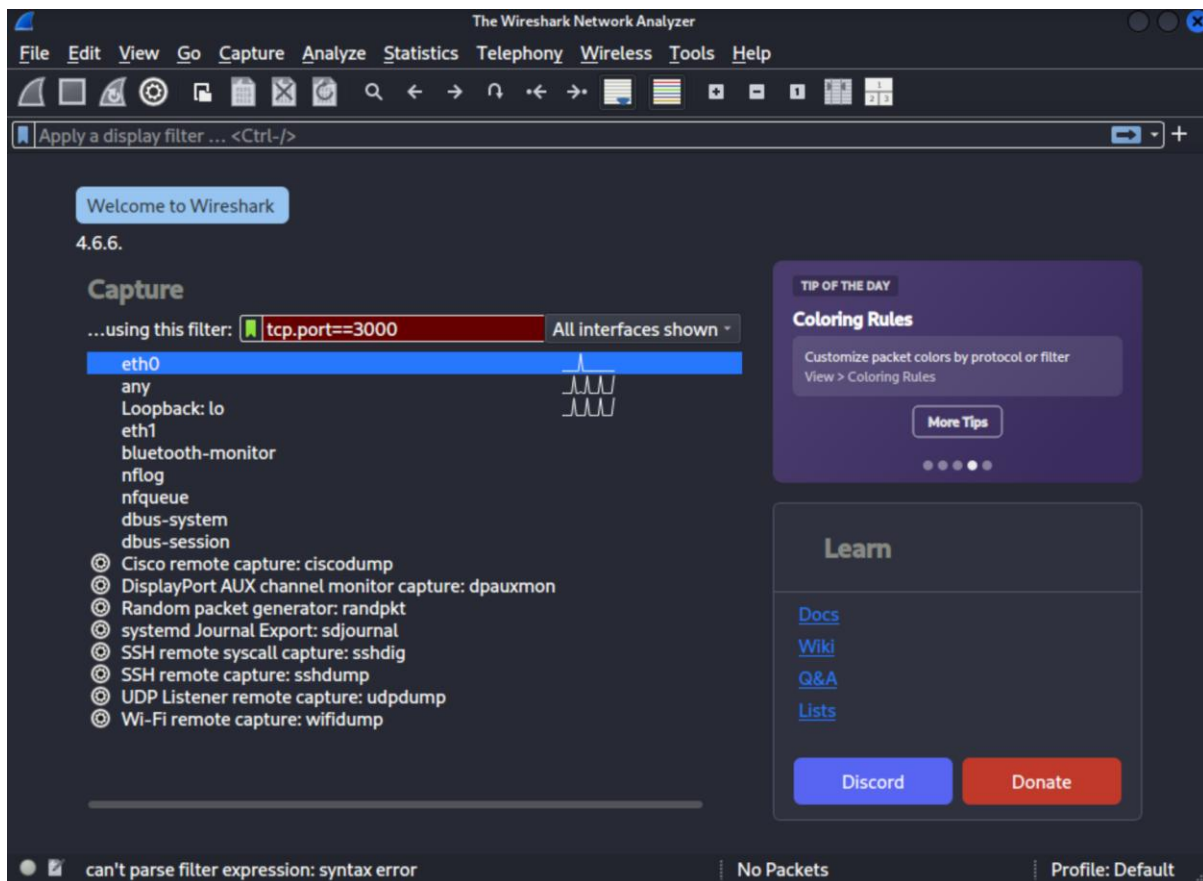
Screenshots:



12. Wireshark

Wireshark captured network packets exchanged between the attacker and target systems. HTTP/TCP traffic was analyzed to validate communication.

Screenshots:



13. Conclusion

This project successfully demonstrated the creation of a personal cybersecurity lab using VMware Fusion. The isolated environment enabled safe deployment of a vulnerable web application and allowed penetration testing using tools such as Nmap, Burp Suite, and Wireshark. The lab provides a strong foundation for future cybersecurity exercises, including vulnerability assessment, web application testing, and network analysis.